BGIN Block#12 Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Blockchain Analysis vs Forensic Analysis

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The Global Network for Blockchain Stakeholders™

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BGIN Block#12

Blockchain Analysis and Forensic Analysis

Date: Monday, March 3, 2025

Location: Tokyo, Japan

Duration: Approx. 90 minutes

Summary

Research Overview

- The initiative originated from a session held in Tokyo about a year ago. Initial discussions aimed to identify a taxonomy of on-chain analysis tools and systems.

- Blockchain analysis companies were hesitant to share information due to its role in their competitive advantage (e.g., tagging addresses). Despite this, there was a shared motivation to promote safety and soundness in the blockchain space. As a result, a formal research proposal was created by BGIN.

- Several workshops were held, with progress made toward a common document. A case study on the Bitcoin Fog case was included, as it exemplifies the difference between analysis and forensics. The Bitcoin Fog case was pivotal in validating blockchain forensics as a legitimate science in court.

- The goal is to establish a clear distinction between blockchain analysis and forensic evidence. This distinction aims to build a common language and shared understanding within the industry and regulatory communities. Blockchain forensics is aligned with legal evidence in court, while analysis is broader and can be used for risk-based decision-making, often in financial services.

- The work is currently in the information-gathering phase. The methodology includes stakeholder interviews with:

    - Blockchain analytics firms
    - Legal practitioners
    - Law enforcement
    - Experts and academic researchers

- An interview template has been prepared to collect primary data. Literature review has been conducted, with a focus on the Bitcoin Fog case. Technical analysis is required to understand what qualifies as forensic evidence vs. general analysis. The team is identifying new stakeholders to support the technical aspects of the paper.

- Expected outputs include:
    - A framework document
    - Possibly a standards proposal
    - Industry guidelines to distinguish between forensic and analytical use cases

- The current effort has been in development for about a year, and the team hopes to complete interviews and integrate findings within the next six months.

<u>Research Goal / Who is the target audience?</u>

- The primary audience for the document is judges, prosecutors, and legal decision-makers, especially those unfamiliar with the technical nuances of blockchain.

- The Bitcoin Fog case illustrated the challenge: in criminal cases, the discussion of technical tools like blockchain analytics vs. forensics often arises unexpectedly in court.

- Judges and prosecutors are often not the ideal parties to be responsible for in-depth technical analysis. Many investigators lack a deep understanding of what blockchain tools do and what their own responsibilities are in interpreting tool outputs.

    - Tool vendors often clarify that tools do not perform investigations, but that message doesn’t always reach end users effectively.

    - The paper would provide clear guidance: what the investigator is expected to understand and explain in court vs. what the tool can and cannot do.

        - The analogy was made to a TI-82 calculator: it can plot graphs but can't do trigonometry for you. The user must understand and explain the math.

- A separate, academic-style explanatory document is needed so courts can reference it outside of the courtroom environment.

    - In the U.S., courts follow the Daubert standard, which governs the admissibility of expert witness testimony and scientific evidence.

    - The concern is that if tool companies themselves publish such material, it could be dismissed as biased or self-serving, even if accurate. A neutral nonprofit organization like BGIN publishing the document would offer more credibility and serve as an independent reference.

    - The document is intended to clarify the distinction between blockchain forensics (evidence) and analysis (interpretation or inference). It would also serve as a training tool for investigators and law enforcement agents.

- The document is envisioned as a practical aid, setting best practices for how to use blockchain analysis tools responsibly and effectively in legal and investigative settings.

- One of the goals of the document should be to act as a universal reference—almost like "Exhibit A"—in blockchain-related financial crime cases.
    - The document could serve as a standard explanatory reference on the difference between forensics and analysis. It would equip defense attorneys with clarity on what they can and cannot ask during expert testimony. It would also equip prosecutors with understanding of how to frame their arguments and extract meaningful testimony from experts. The document would be helpful regardless of whether the expert is an investigator or from a blockchain tool company.
    - Framing the document in this way—i.e., as a courtroom aid—could significantly improve its utility and impact. The idea is to standardize interpretation and expectations around blockchain evidence in legal proceedings.

<u>Paint Points by Different Stakeholders</u>

- One speaker noted that investigators and prosecutors often ask how to use blockchain analysis tools appropriately in court.
    - There is uncertainty and lack of standards, especially in countries like Japan, which is perceived to be behind the U.S. in this area.
    - A recurring question from practitioners is about the boundary between usable data and unsupported inference in an official investigation report.
- Blockchain analytics companies must be extremely cautious about how their data is used, particularly because address identification data is a proprietary business asset. While companies might provide clients with documentation, nuances often require verbal explanation that cannot easily be conveyed in emails or written reports. This lack of clarity and difficulty in conveying nuance is a major challenge for analysis firms.
    - There's an added complexity in explaining address clustering or associations to a jury or judge unfamiliar with blockchain, even if analysts internally understand and agree on the rationale.
    - Conveying why certain wallet clusters are associated or labeled requires more than just showing data. It requires narrative clarity and careful explanation.
    - These challenges reinforce the need for a standardized, neutral resource to explain what blockchain analysis can(not) in forensic or evidential settings.
- US perspective: There is only a small subset of professionals globally who understand blockchain analysis at a nuanced level, even within the U.S. Many state and local law enforcement agencies lack basic exposure to blockchain concepts like clustering or attribution.

- In the U.S., the quality of legal defense (including expert witnesses) can depend on a defendant's financial means, which is problematic. An independent nonprofit guide could serve as a baseline educational resource for judges, prosecutors, and defense attorneys. A nonprofit framework could clearly explain where the responsibility lies between the tool's functionality and the investigator's analysis. Such a guide would help blockchain analytics companies avoid awkward conversations about the misuse of their tools in sales or training contexts.

- Former blockchain analytics professionals often now serve as expert witnesses, helping courts understand basic crypto concepts. Such a resource could reduce dependency on expensive expert witnesses to explain foundational issues like what is a blockchain or clustering. Having neutral definitions would help resolve discrepancies in terminology between companies like Chainalysis and Elliptic.

- The Bitcoin Fog case highlighted the risks of relying on the quality of a single investigator's explanation, even if they're highly experienced.In the U.S. legal system, defense attorneys can ask wide-ranging questions, so training and standards for investigators are crucial.

- The distinction between analysis and forensics is particularly important in the U.S. legal system, where the standards for evidence differ.

    - Forensics is expected to be repeatable, scientific, and objective, aligning with Daubert standards in U.S. courts. Analysis, like recognizing structuring behavior, relies more on subjective interpretation, training, and experience.

    - In the Bitcoin Fog case, this distinction was critical—highlighting the importance of clearly defining what constitutes forensics.

- Japan perspective: In Japan, crypto scams and money laundering are major concerns, and these are usually handled by detective teams, not cyber units. Trainers report that non-cyber investigators often have an "allergy" to technical terms like IP addresses.

    - Analogies, such as comparing blockchain addresses to rental car license plates, help explain unfamiliar concepts. Many investigators encounter crypto only after tracing fiat transfers to cryptocurrency exchanges.

    - There is an absence of Japanese-language training materials on blockchain analytics for practical investigations.Investigators must currently rely on a mix of technical papers, business articles, and blog posts, which makes learning inconsistent and fragmented.

    - There is a clear need for a standard reference or educational resource that combines all relevant aspects in a single, neutral guide.

- Primary users of analytics tools have historically been cybercrime investigators, who tend to be more technically proficient. However, crypto-related crimes now span financial and fraud cases, reaching non-cyber teams unfamiliar with tech.
- Tool companies should help clarify this, but proprietary methods ("secret sauce") complicate disclosure. Nonetheless, **many baseline heuristics and methodologies are commonly shared across tools and can be standardized. Techniques like spend analysis or temporal clustering are widely known and appear in academic literature, so they can be part of a standard. A shared understanding of baseline methods would help investigators, especially in court testimony situations.**

<u>TradFi vs Crypto</u>

- It's difficult to find a TradFi equivalent to blockchain analysis; bank statements are straightforward, while blockchain involves behavioral inference.
- A close TradFi analogy is "structuring" (e.g., depositing $9,900 repeatedly to avoid $10,000 thresholds), which leads to behavioral inferences.
- This kind of typology-based reasoning is similar to clustering in blockchain analysis, where behavior patterns indicate shared ownership or control.

<u>Data Hygiene</u>

- Data hygiene is a key concept and refers to how data is sourced, labeled, and maintained inside blockchain analysis tools.
- Attributions like “this address belongs to a darknet market” may come from purchases, open-source intelligence, or reports—but those aren't visible to tool users.
- Investigators typically can’t explain the backend data sources and should not be asked to do so in court. Investigators can only say “the tool attributed it as such” or “I confirmed with a subpoena,” but not how the tool originally labeled it.
- Courts may have the right to ask about the origin of attributions if they are used in evidence. The tool companies, not the investigators, must be able to explain how an address was labeled and maintain that audit trail.
- The paper shouldn’t detail how attribution is done, but should flag data hygiene as a topic courts and prosecutors must consider. Data hygiene shouldn’t be treated as secret sauce, even if methods are proprietary—the existence of the concept must be understood by courts. The goal is not to expose trade secrets, but to raise awareness of where responsibilities lie and what questions courts are entitled to ask.

<u>Is Blockchain Analysis often used as court evidence?</u>

- In the UK, courts generally don't accept blockchain analytics company attributions as standalone evidence.
- In the US, such attributions may be used, but it's rare — the Bitcoin Fog case was an exceptional example. The Fog case involved a highly unusual situation where only analytics evidence was relied upon.
    - The case was saved by a rare combination of factors:
        - An experienced investigator,
        - A judge with knowledge of crypto,
        - A tool company representative who was also the investigator,
        - And the defense attorney, while aggressive, didn't succeed due to the judge's knowledge.
    - The defense attorney in Fog asked highly inappropriate questions of the tool company representative. If any one of the positive elements had been missing (e.g. judge's knowledge, skilled investigator), the outcome could have been disastrous.
- This situation highlighted the fragility of relying on analytics evidence without standardized context or guidance. The Fog case served as a canary in the coal mine, revealing that better resources and guidance are urgently needed.

<u>Fragility of Relying on Analytics Evidence</u>

- The Bitcoin Fog case was unique and early — but many more similar cases are expected in the future. Without a baseline standard or framework, future cases will be chaotic and inconsistent.
- Outcomes may end up depending solely on which side — prosecution or defense — is more persuasive, rather than on facts or fairness.
- There's a major risk if prosecutors lack internal resources or support from tool companies. Not all blockchain analytics firms have former prosecutors who can guide current prosecutors through evidence preparation and court strategy. Even if such experts exist within companies, legal or policy reasons might prevent them from being accessible. This creates dangerous variability: prosecutors may be left without clarity on what investigators can or cannot say in court.
- In past government practice, prosecutors shared internal resources (e.g., "go-bys") — but access depended on knowing the right people. Without such shared materials, some had to prepare legal documents or evidence summaries using basic internet searches.
- An objective, nonprofit-led resource document, developed in consultation with tool companies, would:
    - Act as a common, neutral reference,
    - Help level the playing field between well-resourced and under-resourced actors,

    - Prevent knowledge asymmetry from leading to unjust outcomes.

- Defense attorneys are likely to continue exploiting gaps in prosecutor and judge knowledge. If prosecutors and judges lack enough background to identify when a question is inappropriate, defense attorneys gain unfair leverage. A standardized resource would enable all parties to identify and correct these situations early.

<u>Perspectives of Blockchain Analytics Companies</u>

- Blockchain analytics companies are extremely cautious about participating in testimonial support for legal cases. These companies involve their legal teams to carefully assess whether participating in a specific case is:

    - Worthwhile from a business perspective,
    - Appropriate based on legal risk,
    - Aligned with internal strategic goals.

- If a case is deemed too complex or risky, the company may decline to participate altogether. The process of preparing for testimony and defending forensic attributions is extremely resource-intensive. These companies do not want to repeat this level of effort for every case.

- The Bitcoin Fog case, though high-profile, required:

    - Extensive internal review,
    - Legal review,
    - Internal audits,
    - A time commitment of months (potentially over six months).

- From the client's side, such legal processes are also very expensive, requiring large budgets.The Bitcoin Fog case involved top-tier investigators and well-funded U.S. organizations, which is not typical for most cases. Even though blockchain analytics firms are revenue-driven businesses, the effort required for courtroom support is not scalable. Therefore, industry-wide standards or references would be helpful:

    - To reduce the burden on individual analytics companies,
    - To prevent them from having to allocate substantial resources for every case,
    - To ensure fairness and efficiency in court proceedings.

- Some components of blockchain analysis are not resource-intensive and can be replicated easily by others using raw blockchain data. There is a general lack of awareness that certain analytical steps can be independently reproduced without relying entirely on proprietary analytics tools.

- Other parts of blockchain forensics do require:

    - Complex methodologies, or
    - Significant computational power,
    - Which may limit reproducibility or accessibility for average investigators.

- A well-crafted reference document could help clarify this distinction, showing the items below. This clarification could reduce over-reliance on analytics companies and empower others to verify simpler claims independently.

    - What aspects are easily reproducible with open data,
    - Versus what aspects require proprietary tools or intensive resources.

<u>Parallel Reconstruction</u>

- Parallel reconstruction—the process of replicating analytic findings without relying on proprietary tools—used to be encouraged as a safe method for courtroom evidence.

- However, as cases grow more complex, parallel reconstruction is becoming increasingly difficult, especially for anything beyond basic money laundering.

- There is concern that:

    - Parallel reconstruction might no longer be feasible in many advanced cases
    - Investigators might be unfairly seen as failing if they cannot perform it, when in reality, the tool provides the necessary speed and scale.

- In the UK, parallel reconstruction is still seen as the preferred or expected method, due to a more conservative stance on using analytics tools as direct evidence. In Canada, it is believed to be a requirement, though details are uncertain.

- The group discussed the importance of: Clarifying the role of parallel reconstruction in different jurisdictions (e.g., required, best practice, optional) and explaining why some countries insist on it, while others (like the US) don't. If an investigator in the US does perform parallel reconstruction, it should be seen as going above and beyond, rather than being expected.

- There's a need to communicate that failure to reconstruct something doesn't mean the investigator is incompetent—it may highlight the value of the tool.

- The group agreed this is a valuable topic, but cautioned against making the document too long or overly academic (i.e., a *treatise*).

<u>Other elements to be put in the paper</u>

- Focusing on non-proprietary content is more likely to get support from analytics companies. Past working group discussions may have overstepped by asking developers to explain proprietary heuristics, which is unlikely to be shared.

- The realistic and constructive approach is to:
    - Stick to basic concepts of what blockchain analysis tools do.
    - Emphasize best practices for using the tools.

- The value for judges isn't in revealing trade secrets, but in clearly distinguishing between: General functionality and capabilities of tools (“this is sauce”) vs proprietary methodologies (“this is secret sauce”).

- It's important to differentiate common heuristics (e.g. co-spend, UTXO) from proprietary heuristics used by individual companies. Proprietary heuristics require testimony from the tool company—not the investigator—because the investigator likely doesn’t know the specific logic behind tool-generated attributions.

- The paper should clarify the boundary of responsibility between: Investigators (can explain usage, but not tool logic) vs Tool companies (must explain how the tool works if challenged).

- The goal of the document is not to provide the answers to technical/legal questions, but to define who is the appropriate person to ask.

- Reference to the Daubert standard (U.S. legal test for expert testimony admissibility):

    - Document could explain when Daubert applies to blockchain analytics tools.
    - Defense attorneys may raise Daubert challenges, and judges may allow hearings even when inappropriate.
    - The paper can help clarify what scenarios merit Daubert consideration, without disclosing proprietary details.

- The Bitcoin Fog case was a landmark example used by tool companies for credibility and marketing. Important, but only one case—not representative of all potential scenarios. There’s a risk of overgeneralization—some might mistakenly believe Fog set universal precedent. Need for the paper to prevent misinterpretation of the scope and implications of past cases (like Bitcoin Fog).

<u>Next Steps</u>

- Next steps involve creating a rough outline of the document if one doesn’t already exist.

- The document will likely be long due to the complexity of the topic, but:

    - It can be structured with clearly defined sections (e.g. *Daubert standard*, *data hygiene*, *permissible questions*).
    - Individual sections can later be extracted as standalone reference materials.

- It’s important that the document not be U.S.-centric, as legal systems differ: Include input and sections relevant to UK, Canada, and possibly Japan. These jurisdictions represent different levels of maturity in dealing with blockchain forensic evidence.